

# Lab Report 3

## PDF and CDF of Continuous Probability Distributions

Roll Number: 226EC5003

### 1. Aim

To generate and plot the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of four continuous probability distributions — Uniform, Exponential, Gaussian (Normal), and Rayleigh — using Python, and to study their statistical behaviour for parameter values derived from the Roll Number 226EC5003.

### 2. Software Used

- Python 3
- NumPy — for numerical array operations
- SciPy (scipy.stats) — for standard PDF/CDF distribution functions
- Matplotlib — for plotting the PDF and CDF curves

### 3. Theory

#### 3.1 Uniform Distribution

A continuous random variable  $X$  is said to be uniformly distributed over an interval  $[a, b]$  if its probability is spread evenly across the interval. Its PDF and CDF are given by:

$$f(x) = 1 / (b - a) , \quad a \leq x \leq b$$

$$F(x) = (x - a) / (b - a) , \quad a \leq x \leq b$$

In this experiment,  $a = 0$  and  $b = 5$ .

#### 3.2 Exponential Distribution

The exponential distribution models the time between independent events occurring at a constant average rate  $\lambda$ . Its PDF and CDF are given by:

$$f(x) = \lambda e^{-\lambda x} , \quad x \geq 0$$

$$F(x) = 1 - e^{-\lambda x} , \quad x \geq 0$$

In this experiment,  $\lambda = 5$ .

#### 3.3 Gaussian (Normal) Distribution

The Gaussian distribution is the most commonly occurring continuous distribution, characterised by its symmetric bell-shaped curve. It is defined by a mean  $\mu$  and standard deviation  $\sigma$ . Its PDF is given by:

$$f(x) = (2\pi\sigma^2)^{-1/2} \cdot \exp [ -(x - \mu)^2 / 2\sigma^2 ]$$

The CDF  $F(x)$  gives the cumulative probability up to  $x$ , and is computed numerically as there is no closed elementary form. In this experiment,  $\mu = 5$  and  $\sigma = 3$ .

### 3.4 Rayleigh Distribution

The Rayleigh distribution is a continuous distribution often used to model the magnitude of a 2-D vector whose components are independent, zero-mean Gaussian variables — for example, the envelope of a multipath fading wireless channel. Its PDF and CDF (scale parameter  $\sigma$ ) are given by:

$$f(x) = (x / \sigma^2) \cdot \exp[-x^2 / 2\sigma^2], \quad x \geq 0$$

$$F(x) = 1 - \exp[-x^2 / 2\sigma^2], \quad x \geq 0$$

In this experiment,  $\sigma = 4$ .

### 4. Source Code

The code below is organised into notebook-style cells, one per distribution, matching how it was run in Jupyter:

```
[1]: Roll_no = "226EC5003"
print("Roll Number:", Roll_no)

Roll Number: 226EC5003
```

```
[2]: import numpy as np
import matplotlib.pyplot as plt

from scipy.stats import uniform, expon, norm, rayleigh
```

```
[3]: # Uniform Distribution

x = np.linspace(0, 5, 1000)
pdf = uniform.pdf(x, loc=0, scale=5) # PDF
cdf = uniform.cdf(x, loc=0, scale=5) # CDF

plt.figure(figsize=(7, 5))
plt.plot(x, pdf)
plt.title("Uniform Distribution - PDF")
plt.xlabel("x")
plt.ylabel("PDF")
plt.grid(True)
plt.show()

plt.figure(figsize=(7, 5))
plt.plot(x, cdf)
plt.title("Uniform Distribution - CDF")
plt.xlabel("x")
plt.ylabel("CDF")
plt.grid(True)
plt.show()
```

```
[4]: # Exponential Distribution

lambda_value = 5
x = np.linspace(0, 3, 1000)
pdf = expon.pdf(x, scale=1/lambda_value) # PDF
cdf = expon.cdf(x, scale=1/lambda_value) # CDF

# PDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, pdf)
plt.title("Exponential Distribution - PDF")
plt.xlabel("x")
plt.ylabel("PDF")
plt.grid(True)
plt.show()

# CDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, cdf)
plt.title("Exponential Distribution - CDF")
plt.xlabel("x")
plt.ylabel("CDF")
plt.grid(True)
plt.show()
```

```
[5]: # Gaussian Distribution

mu = 5
sigma = 3

x = np.linspace(-10, 10, 1000)
pdf = norm.pdf(x, loc=mu, scale=sigma) # PDF
cdf = norm.cdf(x, loc=mu, scale=sigma) # CDF

# PDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, pdf)
plt.title("Gaussian Distribution - PDF")
plt.xlabel("x")
plt.ylabel("PDF")
plt.grid(True)
plt.show()

# CDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, cdf)
plt.title("Gaussian Distribution - CDF")
plt.xlabel("x")
plt.ylabel("CDF")
plt.grid(True)
plt.show()
```

```
[6]: # Rayleigh Distribution

sigma = 4
x = np.linspace(0, 10, 1000)
pdf = rayleigh.pdf(x, scale=sigma) # PDF
cdf = rayleigh.cdf(x, scale=sigma) # CDF

# PDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, pdf)
plt.title("Rayleigh Distribution - PDF")
plt.xlabel("x")
plt.ylabel("PDF")
plt.grid(True)
plt.show()

# CDF Plot
plt.figure(figsize=(7, 5))
plt.plot(x, cdf)
plt.title("Rayleigh Distribution - CDF")
plt.xlabel("x")
plt.ylabel("CDF")
plt.grid(True)
plt.show()
```

## 5. Procedure

- Import the required libraries: numpy, matplotlib.pyplot, and the relevant distributions (uniform, expon, norm, rayleigh) from scipy.stats.
- Define the Roll Number and extract the 6th and 9th digits to be used as distribution parameters.
- For each distribution, define an appropriate range of x-values using np.linspace.
- Compute the PDF using the .pdf() method and the CDF using the .cdf() method of the respective scipy.stats object.
- Plot the PDF against x, with proper title, axis labels, and grid.
- Plot the CDF against x, with proper title, axis labels, and grid.
- Repeat the above steps for the Uniform, Exponential, Gaussian, and Rayleigh distributions.
- Observe and record the shape and behaviour of each PDF and CDF curve.

## 6. Observations

Roll Number: 226EC5003. The parameter values used for each distribution, a long with the resulting PDF and CDF plots, are given below.

| Distribution | Parameter(s) Used                  | Values                |
|--------------|------------------------------------|-----------------------|
| Uniform      | a (min), b (max)                   | a = 0, b = 5          |
| Exponential  | $\lambda$ (rate)                   | $\lambda = 5$         |
| Gaussian     | $\mu$ (mean), $\sigma$ (std. dev.) | $\mu = 5, \sigma = 3$ |
| Rayleigh     | $\sigma$ (scale)                   | $\sigma = 4$          |

Uniform Distribution

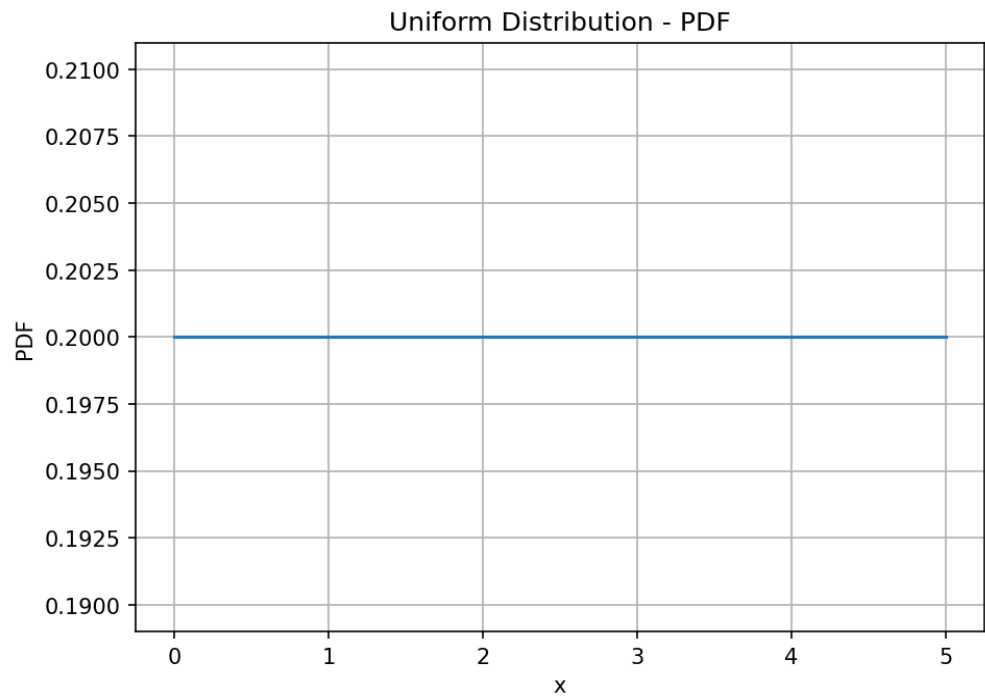


Fig 1: PDF of Uniform Distribution (a = 0, b = 5)

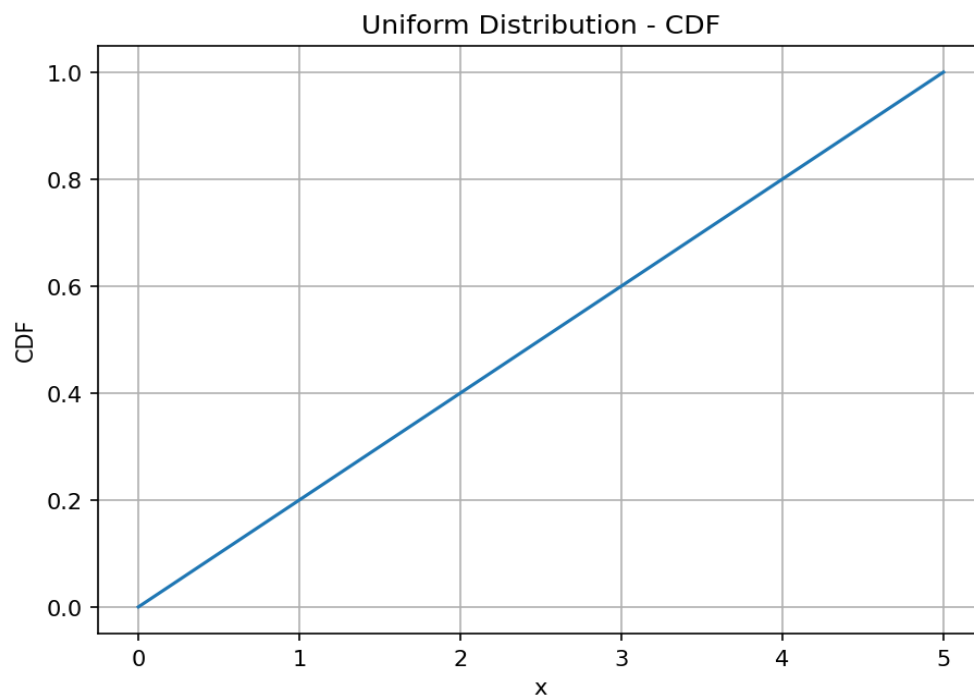


Fig 2: CDF of Uniform Distribution ( $a = 0, b = 5$ )

## Exponential Distribution

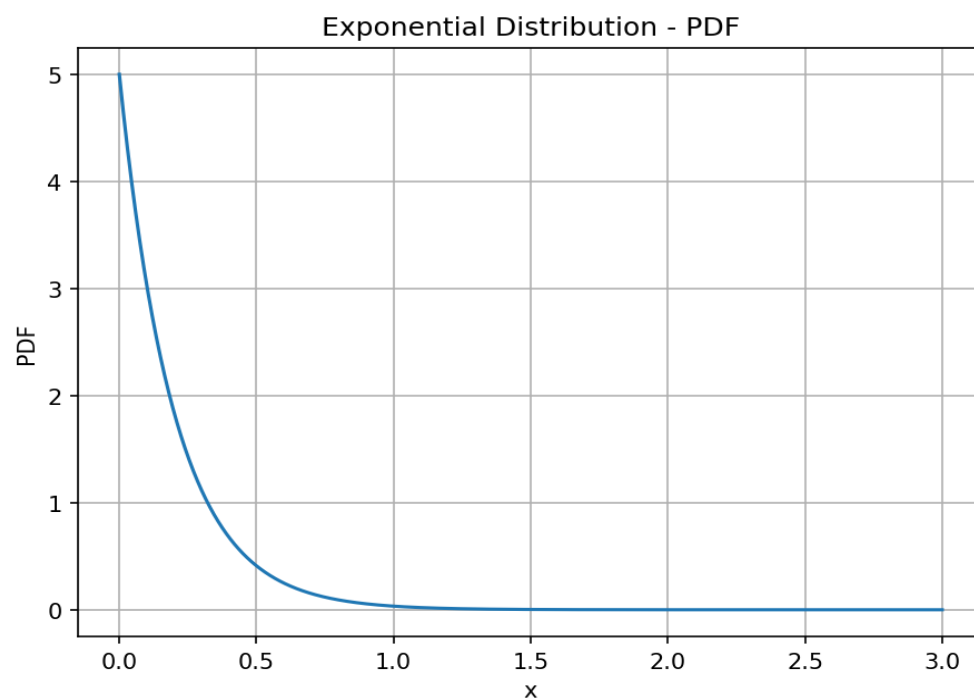


Fig 3: PDF of Exponential Distribution ( $\lambda = 5$ )

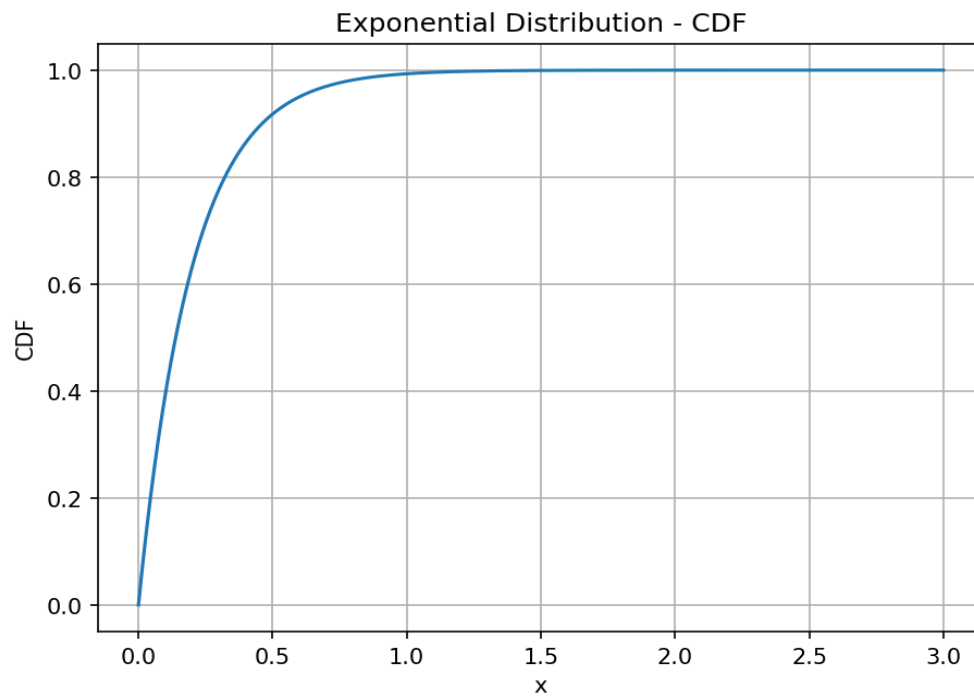


Fig 4: CDF of Exponential Distribution ( $\lambda = 5$ )

## Gaussian Distribution

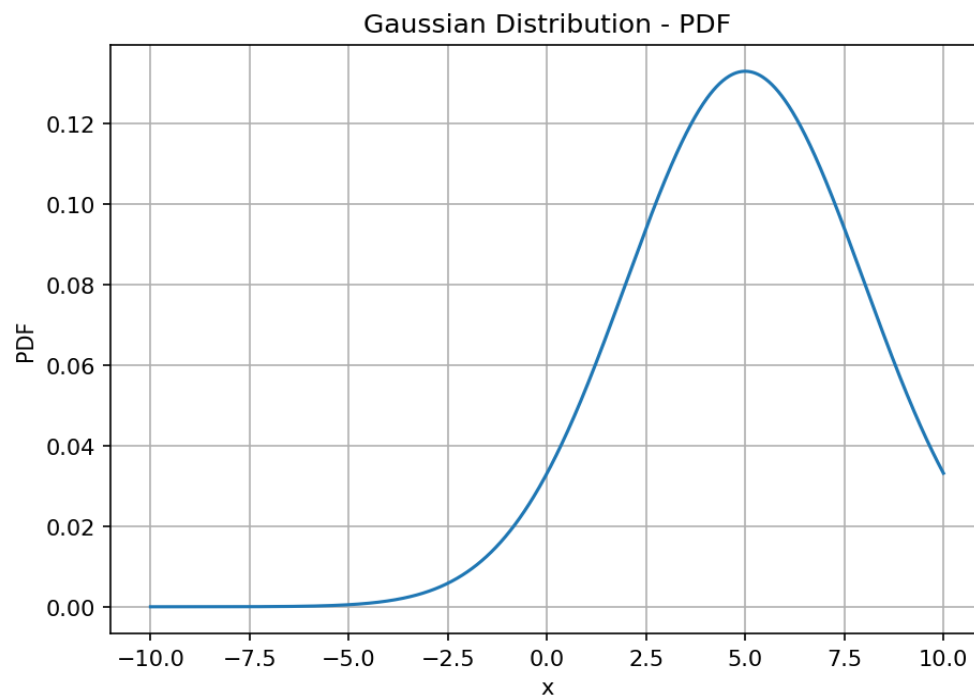


Fig 5: PDF of Gaussian Distribution ( $\mu = 5, \sigma = 3$ )

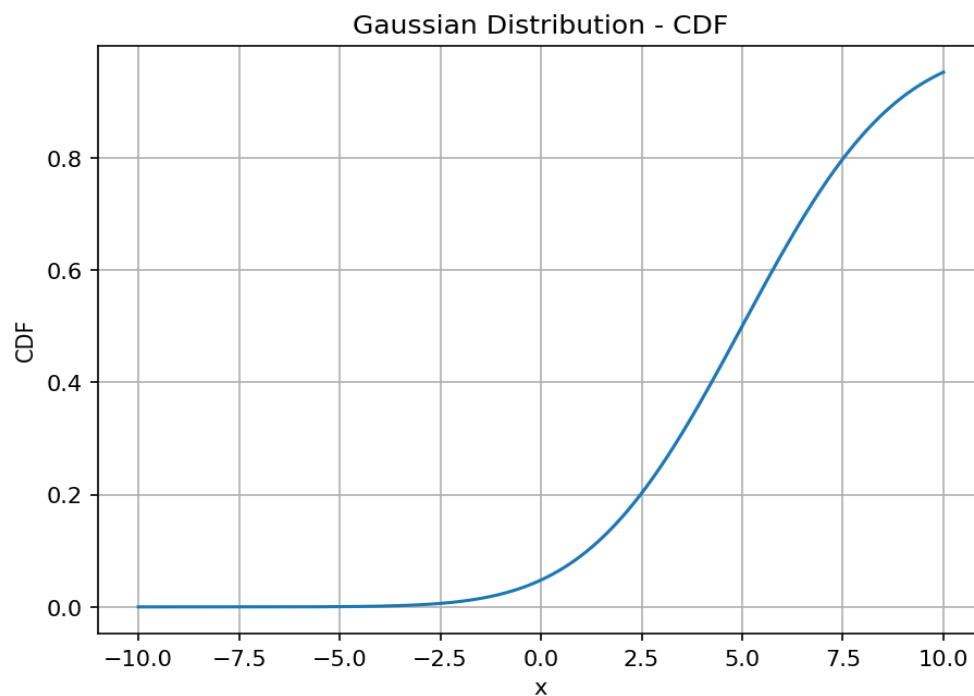


Fig 6: CDF of Gaussian Distribution ( $\mu = 5, \sigma = 3$ )

## Rayleigh Distribution

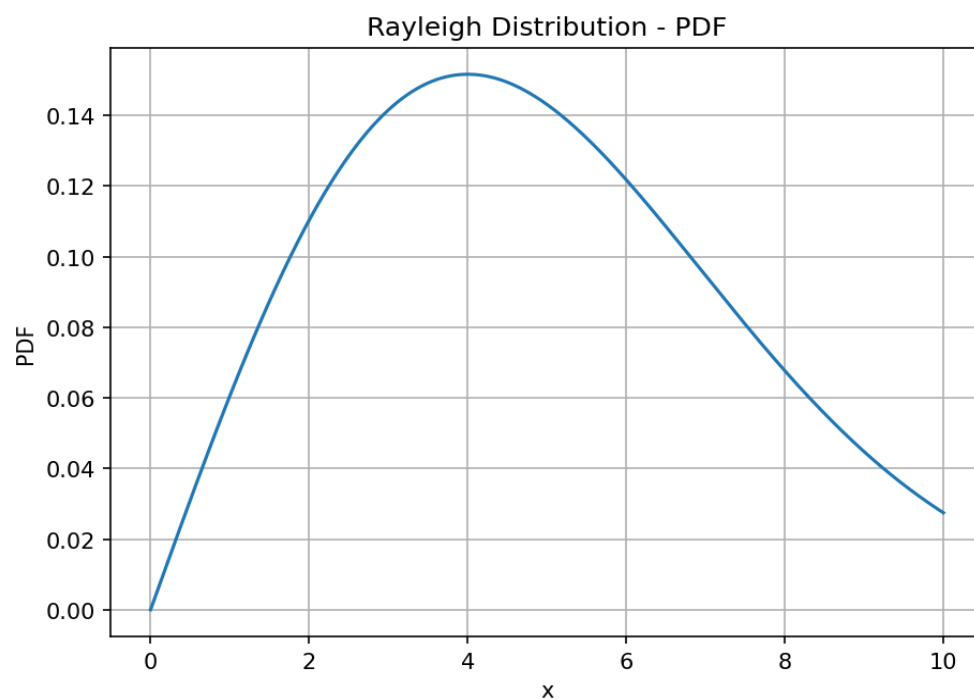


Fig 7: PDF of Rayleigh Distribution ( $\sigma = 4$ )

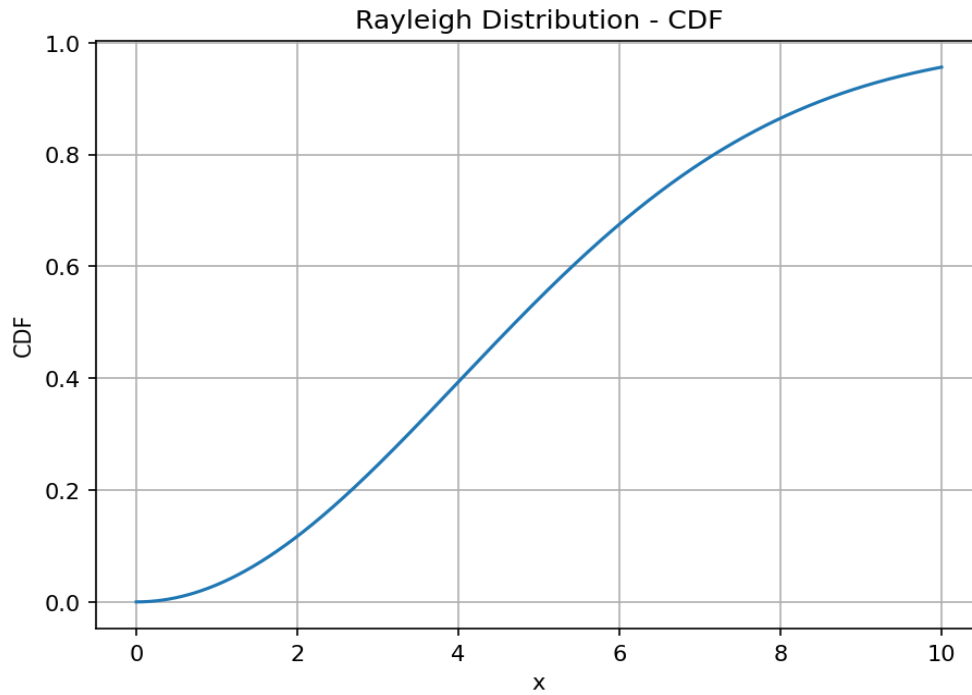


Fig 8: CDF of Rayleigh Distribution ( $\sigma = 4$ )

## 7. Results & Discussion

- The Uniform distribution's PDF is a flat, constant line over  $[0, 5]$  and zero outside it, while its CDF rises linearly from 0 to 1 across the same interval — confirming that probability is spread evenly.
- The Exponential distribution's PDF starts at its maximum value ( $\lambda = 5$ ) at  $x = 0$  and decays rapidly, while its CDF rises quickly and saturates near 1 within a short range, showing that most probability mass is concentrated near  $x = 0$  for a large  $\lambda$ .
- The Gaussian distribution's PDF forms the classic symmetric bell curve centred at  $\mu = 5$  with spread controlled by  $\sigma = 3$ , and its CDF is an S-shaped (sigmoid) curve that crosses 0.5 exactly at  $x = \mu$ .
- The Rayleigh distribution's PDF rises from zero, peaks at  $x = \sigma = 4$ , and then decays, reflecting its use in modelling magnitudes that cannot be negative; its CDF rises monotonically from 0 to 1, matching the skewed shape of the PDF.
- All CDF curves were verified to be non-decreasing and asymptotically approach 1, consistent with the fundamental property of a valid cumulative distribution function.

## 8. Conclusion

The PDF and CDF of the Uniform, Exponential, Gaussian, and Rayleigh continuous probability distributions were successfully generated and plotted in Python using `scipy.stats` and `matplotlib`, with parameter values based on Roll Number 226EC5003. The plots confirm the theoretical shapes expected for each distribution and demonstrate how varying the distribution parameters (range, rate, mean/standard deviation, and scale) affects the spread and shape of both the PDF and the CDF.